

Cover Page

Standard: BACR15CE Rev: (BH) 06-Jan-2006

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ECCN EAR99

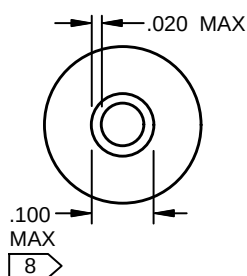
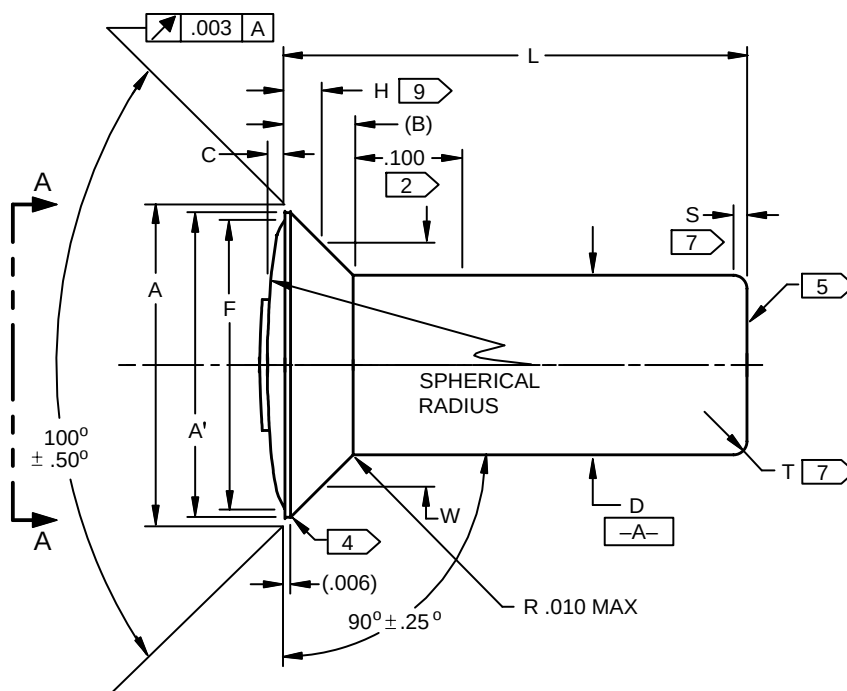
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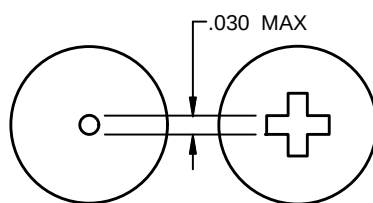
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**FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK**

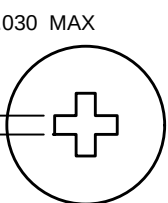
BCA	P	IDS	P	BH	N					
NEW DESIGN APPROVAL: P=PARTIAL, F=FULL, N=NONE										



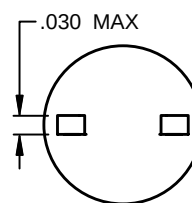
RAISED
CIRCLE
7050 (KE)



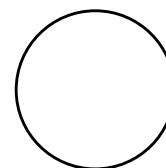
2117 (AD) DIMPLE



5056 (B)
RAISED CROSS



2024 (DD)
RAISED DOUBLE DASH



2017 (D)
NO MARK

VIEW A - A
MATERIAL CODES

DIMENSIONING AND TOLERANCING PER ANSI Y14.5M-1982.
DIMENSIONS IN INCHES UNLESS OTHERWISE NOTED.

TECHNICAL CHANGES IDENTIFIED BY REVISION BAR.

DATE 26-JUN-1956 REV (BH) 06-JAN-2006

CAGE CODE 81205

BACR15CE
SH 1 OF 8

RIVET, 100 DEG SHEAR HEAD

BACR15CE
SH 1 OF 8

***** PSDS GENERATED *****

FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK

TABLE I DIMENSIONS

BOEING STANDARD NUMBER BACR15CE 6	NOM SIZE REF	Ø A TO SHARP CORNER +.000 -.005 1	Ø A' MIN	B HEAD HEIGHT MAX	C +.002 -.001 3	Ø D +.002 -.001 2	Ø F ±.005	H 9		S +.000 -.010	T RAD ±.010	Ø W 9	
								MIN	MAX			MIN	MAX
3	.0938	.148	.129	.024	.003	.093	.091	.0102	.0120	.023	.029	.1190	.1192
4	.1250	.195	.175	.030	.003	.125	.143	.0115	.0133	.030	.039	.1629	.1631
5	.1562	.248	.228	.039	.004	.156	.195	.0136	.0153	.039	.049	.2109	.2111
6	.1875	.298	.279	.048	.005	.187	.203	.0208	.0225	.047	.059	.2438	.2440
8	.2500	.396	.376	.062	.005	.250	.321	.0253	.0269	.062	.078	.3312	.3314

TABLE II DASH NUMBERS FOR STANDARD RIVET LENGTHS 7 10

BOEING STANDARD NUMBER BACR15CE 6	NOM SIZE REF	NOMINAL RIVET LENGTH								
		L ±.010								
		.156	.188	.250	.312	.375	.437	.500	.562	.625
3	.0938	2R5	3	4	5	6	7	8	9	10
4	.1250	2R5	3	4	5	6	7	8	9	10
5	.1562	---	3	4	5	6	7	8	9	10
6	.1875	---	3	4	5	6	7	8	9	10
8	.2500	---	---	---	5	6	7	8	9	10

TABLE II DASH NUMBERS FOR STANDARD RIVET LENGTHS (CONTINUED) 7 10

BOEING STANDARD NUMBER BACR15CE 6	NOM SIZE REF	NOMINAL RIVET LENGTH								
		L ±.010								
		.750	.875	1.000	1.125	1.250	1.375	1.500	1.750	2.000
3	.0938	12	---	---	---	---	---	---	---	---
4	.1250	12	14	16	---	---	---	---	---	---
5	.1562	12	14	16	18	20	22	24	---	---
6	.1875	12	14	16	18	20	22	24	28	---
8	.2500	12	14	16	18	20	22	24	28	32

DATE 26-JUN-1956 REV (BH) 06-JAN-2006

CAGE CODE 81205

BACR15CE

SH 2

RIVET,
100 DEG SHEAR HEAD

BACR15CE

SH 2

***** PSDS GENERATED *****

FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK

NOTES

- 1 "A" DIAMETER TO SHARP CORNER AND HEAD ANGLE ARE SUBJECT TO BOEING INSPECTION BY COMPARATOR METHOD.
- 2 A .001 INCREASE IN "D" DIAMETER IS PERMISSIBLE WITHIN .100 OF HEAD BASE.
- 3 "C" DIMENSION IS THE HEIGHT OF THE CURVED SURFACE ON THE RIVET HEAD. THE SPHERICAL RADIUS MUST BLEND AT DIAMETER "F".
- 4 CURVED EDGE OPTIONAL.
- 5 SEE PART MARKING SECTION.
- 6 SEE CODING SECTION UNDER USAGE AND APPLICATION INFORMATION FOR COMPLETE BOEING PART NUMBER.
- 7 ALUMINUM ALLOY RIVETS BELOW HEAVY LINE IN TABLE II SHALL HAVE RADIUSED ENDS. ALUMINUM ALLOY RIVETS ABOVE HEAVY LINE AND ALL Ni-Cu (MONEL) RIVETS MAY HAVE SQUARE OR RADIUSED END. "S" AND "T" DIMENSIONS APPLY TO RADIUSED END RIVETS ONLY.
- 8 .125 MAXIMUM ON .1562 DIAMETER AND LARGER.
- 9 HEAD PROTRUSION MEASUREMENT IN ACCORDANCE WITH D-11805, INSPECTION METHOD A.
- 10 LONGER NONSTANDARD LENGTHS NOT LISTED IN TABLE II MAY BE ORDERED IN .0625 INCREMENTS BY USE OF THE APPROPRIATE DASH NUMBERS. ADD SUFFIX "R5" TO SECOND DASH NUMBER FOR A NOMINAL LENGTH .03125 LONGER THAN REQUIRED BY THE BASIC DASH NUMBER.
- 11 2024-T4 RIVETS, MATERIAL CODE "DD", REQUIRE RESOLUTION HEAT TREATMENT AND COLD (ICE BOX) STORAGE PRIOR TO INSTALLATION AND THUS CANNOT BE USED IN KITS WHERE THEY ARE SHIPPED AS LOOSE PARTS.
- 12 WITH THE EXCEPTION OF ALLOY 2024, PREVIOUS REVISIONS OF THIS STANDARD PERMITTED EITHER ANODIZE OR CONVERSION COAT FINISH FOR ALUMINUM ALLOYS UNLESS CODE "C" WAS SPECIFIED. CODE "C" RESTRICTED THE FINISH TO CONVERSION COAT ONLY. WITH THE SAME EXCEPTION, FINISH IS NOW LIMITED TO CONVERSION COAT, IDENTIFIED BY THE LETTER "C" ON THE TERMINAL (SHANK) END OF THE RIVET, AND SPECIFIED IN THE PART NUMBER AS THE "NO CODE". FINISH FOR ALLOY 2024 IS ANODIZE. SEE PART MARKING AND CODING SECTIONS.
- 13 BRILES RIVET CORPORATION IS NO LONGER IN BUSINESS. RIVETS MANUFACTURED BY THE BRILES RIVET CORPORATION PRIOR TO OCTOBER 1, 1999 MAY BE RECEIVED UNTIL OCTOBER 1, 2004 PROVIDED THE ORIGINAL BRILES RIVET CORPORATION DATA CERTIFICATIONS ACCOMPANY ALL SHIPMENTS WITH A BRILES RIVET (ALLFAST FASTENING SYSTEMS INC.) CERTIFICATION OF COMPLIANCE.

DATE 26-JUN-1956 REV (BH) 06-JAN-2006

CAGE CODE 81205

BACr15CE

SH 3

RIVET,
100 DEG SHEAR HEAD

BACr15CE

SH 3

***** PSDS GENERATED *****

FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK

NOTES (CONTINUED)

- 14 AEROSPACE RIVET MANUFACTURERS CORPORATION IS NO LONGER IN BUSINESS. RIVETS MANUFACTURED BY THE AEROSPACE RIVET MANUFACTURERS CORPORATION PRIOR TO MAY 1, 2002 MAY BE RECEIVED UNTIL MAY 1, 2006 PROVIDED THE ORIGINAL AEROSPACE RIVET MANUFACTURERS CORPORATION DATA CERTIFICATIONS ACCOMPANY ALL SHIPMENTS WITH AN AEROSPACE RIVET MANUFACTURERS CORPORATION (ALLFAST FASTENING SYSTEMS INC.) CERTIFICATE OF COMPLIANCE.

PROCUREMENT SPECIFICATION

BPS-R-131, EXCEPT AS NOTED.

Ni-Cu (MONEL) RIVETS SHALL BE TESTED TO THE HEADING REQUIREMENTS AND SAMPLING TABLES OF BACR15DX.

MATERIAL

- ALUMINUM ALLOY - 5056-H32, 2017-T4, 2024-T4, 2117-T4 OR 7050-T73 PER QQ-A-430 EXCEPT 2017-H15 WIRE OR ROD MAY BE USED AS RAW MATERIAL FOR 2017-T4 RIVETS.
- Ni-Cu (MONEL) - QQ-N-281, CLASS A, ANNEALED.

HEAT TREATMENT

ALUMINUM ALLOY - PER BPS-R-131.

FINISH

2017-T4, 2117-T4, 5056-H32 AND 7050-T73 ALUMINUM ALLOYS SHALL BE CHEMICAL CONVERSION COATED PER MIL-C-5541, CLASS 1A, CLEAR, COLORLESS. 12

2024-T4 ALUMINUM ALLOY SHALL BE ANODIZED PER MIL-A-8625, CLASS 1, TYPE I, TYPE IB OR TYPE II.

Ni-Cu (MONEL): NO FINISH.

ALUMINUM ALLOY RIVETS SHALL BE CAPABLE OF PASSING SALT SPRAY TESTS PER MIL-STD-1312-1 (NASM1312-1). TEST DURATION SHALL BE 48 HOURS MINIMUM FOR CONVERSION COATED PARTS AND 96 HOURS MINIMUM FOR ANODIZED PARTS. AT THE CONCLUSION OF THE SALT SPRAY TEST THERE SHALL BE NO VISIBLE CORROSION WHEN EXAMINED AT 10X MAGNIFICATION. SAMPLING SHALL BE IN ACCORDANCE WITH ANSI/ASQC Z1.4, INSPECTION LEVEL S3, AQL EQUIVALENT OF 6.5 PERCENT DEFECTIVE. LOT SIZE SHALL BE DEFINED IN TERMS OF POUNDS AND SAMPLING SHALL BE DEFINED IN TERMS OF NUMBERS OF RIVETS.

DATE 26-JUN-1956 REV (BH) 06-JAN-2006

CAGE CODE 81205

BACR15CE**SH 4**

**RIVET,
100 DEG SHEAR HEAD**

BACR15CE**SH 4**

***** PSDS GENERATED *****

FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK

MARKING**MATERIAL IDENTIFICATION ON HEAD**

IDENTIFY ALUMINUM ALLOY RIVETS IN ACCORDANCE WITH THE GRAPHIC.
MARKING SHALL BE RAISED OR DEPRESSED .010 MAXIMUM, AS APPLICABLE.

Ni-Cu (MONEL) RIVETS HAVE NO MATERIAL IDENTIFICATION MARKING.

IDENTIFICATION OF CONVERSION COATED ALUMINUM RIVETS

CONVERSION COATED RIVETS MANUFACTURED AFTER NOVEMBER 1, 1995 SHALL BE MARKED WITH THE LETTER "C" ON THE TERMINAL (SHANK) END, RAISED OR DEPRESSED .020 MAXIMUM, SIZE OPTIONAL. "C" MARKING IS OPTIONAL ON RIVETS MANUFACTURED PRIOR TO THAT DATE.

CONVERSION COATED RIVETS WITHOUT THE "C" MARKING MAY BE RECEIVED BY BOEING AND THEIR SUBCONTRACTORS UNTIL NOVEMBER 1, 1997 PROVIDED THAT THE RIVETS WERE MANUFACTURED PRIOR TO NOVEMBER 1, 1995. EXISTING STOCKS OF RIVETS WITHOUT THE "C" MARKING MAY BE USED UNTIL DEPLETED.

PACKAGING

AFTER AUGUST 1, 1995, THE FOLLOWING PACKAGING REQUIREMENTS ARE MANDATORY FOR ALL MATERIALS EXCEPT ALLOY 2024.

RIVETS OF THE SAME LOT, STYLE, SIZE AND PART NUMBER SHALL BE PACKED IN ONE POUND SEALED PLASTIC BAGS. THE ONE POUND BAGS SHALL BE PACKED IN INTERMEDIATE OR UNIT CONTAINERS (RIVET BOX) WHICH CONTAIN A MAXIMUM NET WEIGHT OF TEN POUNDS. FINAL PACKAGING SHALL BE IN ACCORDANCE WITH BPS-R-131.

THE FOLLOWING INFORMATION SHALL BE MARKED ON THE OUTSIDE OF EACH ONE POUND BAG AND EACH INTERMEDIATE OR UNIT CONTAINER:

- a. DATE OF MANUFACTURE OR HEAT TREAT.
- b. MANUFACTURER'S LOT NUMBER.
- c. BOEING PART NUMBER.
- d. NAME OF MANUFACTURER.
- e. BAR CODE WHEN AND AS SPECIFIED ON PURCHASE ORDER.

ALLOY 2024 RIVETS SHALL BE PACKAGED IN ACCORDANCE WITH BPS-R-131.

NOTE: PACKAGING REQUIREMENTS FOR ALLOY 2024 ARE DIFFERENT BECAUSE 2024 IS RESOLUTION HEAT TREATED PRIOR TO PRODUCTION USE.

DATE 26-JUN-1956 REV (BH) 06-JAN-2006

CAGE CODE 81205

BACr15CE**SH 5**

**RIVET,
100 DEG SHEAR HEAD**

BACr15CE**SH 5**

***** PSDS GENERATED *****

FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK**COLOR**

NATURAL.

PROCUREMENTAEROSPACE RIVET MANUFACTURERS CORP (CAGE CODE 64728) 

ALLFAST FASTENING SYSTEMS INC (CAGE CODE 53551)

ATELIERS DE LA HAUTE-GARONNE (AHG) (CAGE CODE F0095)

BRILES RIVET CORP (CAGE CODE 55580) 

THE MANUFACTURERS LISTED IN BPS-R-131SUP AND THEIR AUTHORIZED DISTRIBUTORS ARE THE ONLY APPROVED SOURCES FOR THE ABOVE QUALIFIED PRODUCTS. SEE BPS-R-131SUP FOR PLANT ADDRESSES. NO CHANGES IN PRODUCT DESIGN, BASIC METHODS OF MANUFACTURE, PLANT SITE OR QUALITY LEVEL SHALL BE MADE WITHOUT PRIOR NOTIFICATION AND PRIOR APPROVAL IN WRITING FROM THE BOEING COMPANY. MANUFACTURERS OF COMPETITIVE PRODUCTS MAY APPLY TO A SUPPLIER MANAGEMENT AND PROCUREMENT DEPARTMENT OF THE BOEING COMPANY FOR QUALIFICATION. IF A MANUFACTURER IS SHOWN ON THIS STANDARD, BUT NOT LISTED IN THE SUPPLEMENT, CONTACT THE DIVISIONAL ENGINEERING STANDARDS FOCAL POINT OR ENGINEERING STANDARDS FOR VERIFICATION.

USAGE AND APPLICATION INFORMATION INSTALL PER BAC5004-1, BAC5058 OR BAC5063-1, AS APPLICABLE.

SEE BACR15DS AND BACR15EA FOR OVERSIZE RIVETS.

DATE 26-JUN-1956 REV (BH) 06-JAN-2006

CAGE CODE 81205

BACR15CE

SH 6

**RIVET,
100 DEG SHEAR HEAD****BACR15CE**

SH 6

***** PSDS GENERATED *****

FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK

CODING

FIRST DASH NUMBER DESIGNATES NOMINAL DIAMETER IN .03125 INCREMENTS.

CODE "AD" DESIGNATES 2117-T4 MATERIAL

CODE "B" DESIGNATES 5056-H32 MATERIAL

CODE "D" DESIGNATES 2017-T4 MATERIAL

CODE "DD" DESIGNATES 2024-T4 MATERIAL 11

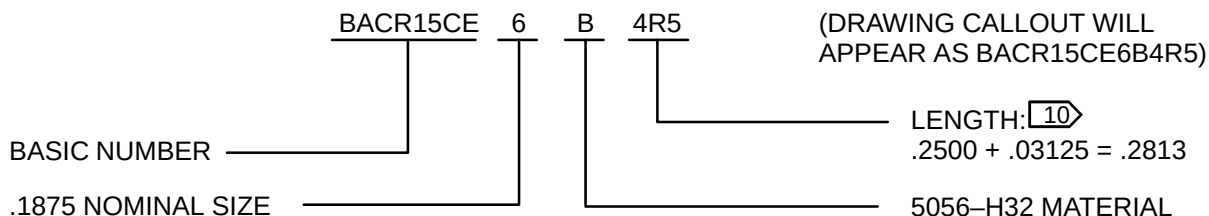
CODE "M" DESIGNATES Ni-Cu (MONEL) MATERIAL

CODE "KE" DESIGNATES 7050-T73 ALUMINUM ALLOY

SECOND DASH NUMBER DESIGNATES NOMINAL LENGTH IN .0625 INCREMENTS

CODE "R5" FOLLOWING SECOND DASH NUMBER DESIGNATES A NOMINAL LENGTH
.03125 LONGER THAN THE BASIC INCREMENT LENGTHCODE "C" IS AN OBSOLETE DESIGNATION FOR CHEMICAL CONVERSION COAT FINISH.
CONVERSION COAT IS NOW THE ONLY PERMITTED FINISH FOR BACR15CE RIVETS
OTHER THAN ALLOY 2024, AND IS SPECIFIED AS THE "NO CODE". 12

EXAMPLE OF PART NUMBER



DATE 26-JUN-1956 REV (BH) 06-JAN-2006

CAGE CODE 81205

BACR15CE

SH 7

**RIVET,
100 DEG SHEAR HEAD****BACR15CE**

SH 7

***** PSDS GENERATED *****

FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK

FASTENER CODE

SEE BACD2074 FOR FASTENER CODES.

SEE D-590-PREFACE (INDEX) FOR INACTIVATION DEFINITIONS. SEE
D-590-SUPERSESSION-LIST FOR SUPERSESSION CLASS DEFINITIONS AND SUPERSESSION
LIST.

INACTIVATION APPLICABILITY

- BCA, IDS -

BACR15CE3()2.5 AND BACR15CE4()2.5 ARE INACTIVE FOR DESIGN AND
PROCUREMENT. BACR15CE3()2R5 AND BACR15CE4()2R5
RESPECTIVELY ARE CLASS II SUPERSESSIONS.

BACR15CE()(B,D,AD,DD,KE)()C RIVETS ARE INACTIVE FOR DESIGN AND
PROCUREMENT. BACR15CE()(B,D,AD,DD,KE)() (NO CODE) RIVETS ARE
CLASS II SUPERSESSIONS. SEE MARKING SECTION FOR EFFECTIVITY OF
"C" MARKING REQUIREMENTS.
- BCA -

BACR15CE()D() RIVETS ARE INACTIVE FOR DESIGN AND PROCUREMENT.
BACR15GF()D() RIVETS ARE CLASS II SUPERSESSIONS.

DATE 26-JUN-1956 REV (BH) 06-JAN-2006

CAGE CODE 81205

BACR15CE

SH 8

RIVET,
100 DEG SHEAR HEAD

BACr15CE

SH 8